

CURRICULUM VITAE

Boqiang XU

Tel: (+86) 15801739930; (+852) 44602404

Email: bq.xu@cityu.edu.hk; xubq628@gmail.com

Citations at Google Scholar = 331 with h-index = 9 (As of July 2026)

Nationality: Chinese

Education

Tongji University (Shanghai, China)

- *Department of Bridge Engineering, College of Civil Engineering* **Mar. 2021 - Aug. 2025**
Ph.D. Candidate of civil engineering (integrated master's and doctoral program)

National University of Singapore (Singapore)

- *Department of the Built Environment, College of Design and Engineering* **Nov. 2023 – Oct. 2024**
Visiting Ph.D. Student

Tongji University (Shanghai, China)

- *Department of Bridge Engineering, College of Civil Engineering* **Sep. 2019 – Jan. 2021**
Master Candidate of civil engineering (exempt from Admission exam)

Tongji University (Shanghai, China)

- *College of Civil Engineering* **Sep. 2015 – Jun. 2019**
Bachelor of Civil Engineering
- *College of Mathematical Sciences* **Sep. 2015 – Jan. 2017**
Minor in Mathematics and Applied Mathematics

Experience

City University of Hong Kong (Hong Kong SAR, China)

- *Department of Architecture and Civil Engineering* **Nov. 2025 - Present**
Postdoctoral Fellow

Research Interests

Structural health monitoring, Digital twin, Computer vision, Machine/Deep learning, Signal processing

- Application of monocular 3D vision algorithms based on geometric prior information (standardized 3D reconstruction, automated homography estimation).
- Vision detection algorithms for structural or bridge damages under small samples or changing illumination.
- Vision system for accurate spatial-temporal distribution identification of vehicle loads in complex environments.
- Time-frequency feature analysis and blind source separation of sensor data in bridge health monitoring.
- Integration of Bayesian methods and deep learning in time-series prediction models for sensor data.
- Vehicle load inversion and structural condition assessment based on vehicle-induced structural responses.

Main Research Experience

Research on Intelligent Operation and Maintenance Technology of Long-Span Hybrid Beam Rigid Frame Bridge Based on Vehicle-Bridge Correlation Prediction **Key Researcher**
Jun. 2022 – Sep. 2024

Science and Technology Project of Zhejiang Provincial Department of Transportation, China

- Deep learning framework for real-time vehicle axle load position detection combined with a weigh-in-motion (WIM) system.
- Homography matrix estimation algorithm for camera-to-road based on lane geometric prior information, eliminating the requirement for on-site labeling.

- GAN model for detecting road/bridge cracks under small sample scenarios.
- Nighttime crack detection method based on image-to-image translation models.
- Weakly-supervised and unsupervised algorithms for damage detection in scenarios with limited labeled data.

Research on Intelligent Vehicle Bridge Correlation Monitoring Technology of Long-Span Continuous Beam Bridge with Corrugated Steel Webs Key Researcher
Aug. 2022 – Aug. 2024

Science and Technology Project of Hebei Provincial Department of Transportation, China

- Monocular 3D inference algorithm integrated with deep learning models for identifying the spatial-temporal distribution of vehicle wheel loads.
- Vehicle wheel load identification framework under dense heavy flow using image inpainting models.
- Bridge strain high-frequency information extraction algorithms based on wavelet multiresolution analysis.
- Deep kernel regression Bayesian model for time-series forecasting of temperature-induced bridge strain.
- Correlation analysis of vehicle load-bridge response and development of a B-WIM for bridges.

Honours and Scholarships

2025	The Excellent Ph.D. Graduate, Tongji University
2025	Silver Award of China International College Students' Innovation Competition, Ministry of Education, China
2023	Visiting Ph.D. Program Scholarship of China Scholarship Council, Ministry of Education, China
2022, 2024	Scholarship of Excellence for Doctoral Students, Tongji University
2021	Scholarship of Excellence for Doctoral Freshmen, Tongji University
2020	Second Prize of the 13th China Postgraduate Mathematical Modeling Competition, Association of Chinese Graduate Education
2020	Scholarship of Excellence for Postgraduates, Tongji University
2016	First Prize in the 8th Mathematics competition of Chinese College Students, Chinese Mathematical Society
2016~2018	Scholarship of Excellence for Undergraduates, Tongji University

Publications

Journal Articles:

Boqiang XU, Chao LIU. [A 3D reconstruction method for buildings based on monocular vision](#). *Computer-Aided Civil and Infrastructure Engineering*, 2022, 37(3): 354-369.

Boqiang XU, Chao LIU. [Pavement crack detection algorithm based on generative adversarial network and convolutional neural network under small samples](#). *Measurement*, 2022, 196: 111219.

Chao LIU, **Boqiang XU**. [A night pavement crack detection method based on image-to-image translation](#). *Computer-Aided Civil and Infrastructure Engineering*, 2022, 37(13): 1737-1753.

Boqiang XU, Xingbao LIU, Genyu FENG, Chao LIU. [A monocular-based framework for accurate identification of spatial-temporal distribution of vehicle wheel loads under occlusion scenarios](#). *Engineering Applications of Artificial Intelligence*, 2024, 133: 107972.

Boqiang XU, Chao LIU. [Keypoint detection-based and multi-deep learning model integrated method for identifying vehicle axle load spatial-temporal distribution](#). *Advanced Engineering Informatics*, 2024, 62: 102688.

Boqiang XU, Chao LIU. [A deep kernel regression-based forecasting framework for temperature-induced strain in large-span bridges](#). *Engineering Structures*, 2025, 323:119259.

Boqiang XU, Chao LIU. [Bayesian-optimized deep learning model for real-time spatial distribution identification of vehicle axle load](#). *Expert Systems with Applications*, 2024, 21: 12572.

Boqiang XU, Xinkai Jiao, Qian-Cheng Wang, Chao LIU. [A review of computer vision-based algorithms for vehicle](#)

[load spatial-temporal distribution identification](#). *Measurement Science and Technology*, 2026.

Qixin SUN, Ruiping ZHENG, **Boqiang XU**. [Bayesian-optimized unsupervised semantic segmentation model for structural crack detection](#). *Advances in Structural Engineering*, 2024, 27 (3), 432-449.

Tao WANG, Mingkai LI, Hanmo WANG, Peibo LI, **Boqiang XU**, Difeng HU. [Context-aware depth estimation for improved 3D reconstruction of homogeneous indoor environments](#). *Automation in Construction*, 2025, 177:106343.

Chao LIU, **Boqiang XU**. [Weakly-supervised structural surface crack detection algorithm based on class activation map and superpixel segmentation](#). *Advances in Bridge Engineering*, 2023, 4 (1), 27.

Yixun WU, Taiyu WANG, Runze GU, Chao LIU, **Boqiang XU**. [Nighttime vehicle detection algorithm based on image translation technology](#). *Journal of Intelligent & Fuzzy Systems*, 2024, 46 (2): 5377-5389.

Conference Articles:

Boqiang XU, Chao LIU. [Detection algorithm of structural surface cracks based on class activation map](#). *International Association for Bridge and Structural Engineering (IABSE) Congress*, Nanjing, China, 2022.

Boqiang XU, Zhijun SHI, Chao LIU. [Real-time traffic load monitoring framework based on deep learning model and statistical regularities of vehicle shape prior information](#). *The Eighteenth East Asia-Pacific Conference on Structural Engineering and Construction (EASEC-18)*, Chiang Mai, Thailand, 2024.

Boqiang XU, Xiuzhen TIAN, Junhui HAN, Chao LIU, Qian-Cheng WANG. [A Two-Stage Blind Source Separation Method for Structural Health Monitoring Signals Integrating Deep Kernel Regression and Wavelet Multiresolution Analysis](#). *43rd International Symposium on Automation and Robotics in Construction (ISARC2026)*, Singapore, 2026.

Patent

Chao Liu, **Boqiang XU**. A method for calibrating external parameters of a zoom camera for 3D reconstruction of buildings. Invention Patent: CN202010431827.3, 2023-05-02.

Chao Liu, **Boqiang XU**. A real-time vehicle position detection method based on road geometric information. Invention Patent: CN202110641118.2, 2023-02-10.

Chao Liu, **Boqiang XU**. A monitoring method for vehicle axle load position. Invention Patent: CN115661749B, 2026-03-24.

Xu Jiang, Ziqi Xu, Ao Wang, Zhenbang He, **Boqiang XU**, Chengzhi Yang, Chenyu Hao. A beam segment sliding device suitable for curved beams. Invention Patent: CN201811427336.0, 2020-06-02.

Software Copyright

Chao Liu, **Boqiang XU**, et al. Crack Recognition and Detection Software in Complex Environments. Tongji University, September 2021, Registration No.: 2021SR1353912.

Chao Liu, **Boqiang XU**, et al. Concrete Multi-Scale Crack Recognition Software. Tongji University, August 2022, Registration No.: 2022SR1261160.

Chao Liu, **Boqiang XU**, et al. Nighttime Structural Surface Defect Detection Software. Tongji University, April 2022, Registration No.: 2022SR0531677.

ACADEMIC SERVICE

Reviewer of Journals:

- *Expert Systems with Applications* | *Structural Control and Health Monitoring* | *The Journal of Supercomputing* | *Scientific Reports* | *Discover Applied Sciences* | *Discover Artificial Intelligence* | *Journal of Civil Structural Health Monitoring*

Guest Editor:

- *Frontiers in Built Environment*, “Advancements in Multi-Scale Sensing, Simulation, and Prediction of Urban Resilience from Infrastructure to City Level”
- *Computational Urban Science*, “Human-Centered Urban Digital Twins and Policy Insights”